TimeLens: Rethinking Video Temporal Grounding with Multimodal LLMs

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https://timelens-arc-lab.github.io/

Abstract

*This paper does not introduce a novel method but instead establishes a straightforward, incremental, yet essential baseline for video temporal grounding (VTG), a core capability in video understanding. While multimodal large language models (MLLMs) excel at various video understanding tasks, the recipes for optimizing them for VTG remain under-explored. In this paper, we present **TimeLens**, a systematic investigation into building MLLMs with strong VTG ability, along two primary dimensions: data quality and algorithmic design. We first expose critical quality issues in existing VTG benchmarks and introduce **TimeLens-Bench**, comprising meticulously re-annotated versions of three popular benchmarks with strict quality criteria. Our analysis reveals dramatic model re-rankings compared to legacy benchmarks, confirming the unreliability of prior evaluation standards. We also address noisy training data through an automated re-annotation pipeline, yielding **TimeLens-100K**, a large-scale, high-quality training dataset. Building on our data foundation, we conduct in-depth explorations of algorithmic design principles, yielding a series of meaningful insights and effective yet efficient practices. These include interleaved textual encoding for time representation, a thinking-free reinforcement learning with verifiable rewards (RLVR) approach as the training paradigm, and carefully designed recipes for RLVR training. These efforts culminate in **TimeLens models**, a family of MLLMs with state-of-the-art VTG performance among open-source models and even surpass proprietary models such as GPT-5 and Gemini-2.5-Flash. All codes, data, and models will be released to facilitate future research.*

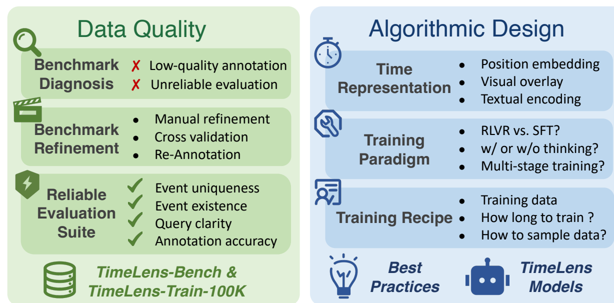

Figure 1. **Overview of the proposed TimeLens framework**. We systematically explore the key factors for building performant video temporal grounding models, dissecting our efforts along two primary dimensions: data quality and algorithmic design. For data quality, we focus on benchmark diagnosis, benchmark refinement, and creating a reliable evaluation suite. For algorithmic design, we study various aspects including time encoding, training recipes, and optimization strategies to establish best practices and develop the TimeLens models.

1. Introduction

Recent multimodal large language models (MLLMs) have excelled at understanding "what" happens in a video, yet they largely fail when asked "when." This limitation is central to the task of video temporal grounding (VTG). The challenge is twofold: 1) VTG necessitates a fundamental shift from coarse semantic aggregation to fine-grained time-aware perception; 2) Distinguishing queried events requires modeling long-term visual dynamics over appearance-centric features, which are notoriously difficult to annotate and learn. As MLLMs become integral to perception [43, 44, 55, 58] and reasoning systems [6, 13, 37, 39, 40, 66], equipping them with robust temporal awareness is no longer optional, but essential [26, 35, 46, 49, 54].

This work focuses on post-training MLLMs with leading temporal grounding ability. This investigation is a straightforward extension given the recent progress in pretrained foundation MLLMs [2, 3, 53]. Different from heavily studied general understanding tasks, recipes for fine-grained grounding tasks are not yet to be established. This paper aims to systematically investigate core components of building time-aware MLLMs (Fig. 1) along two primary dimensions: **data quality** and **algorithmic design**.

Our investigation starts by exposing critical flaws in evaluation benchmarks. We find that existing VTG benchmarks [11, 25, 27] not only lack a clear comparison between

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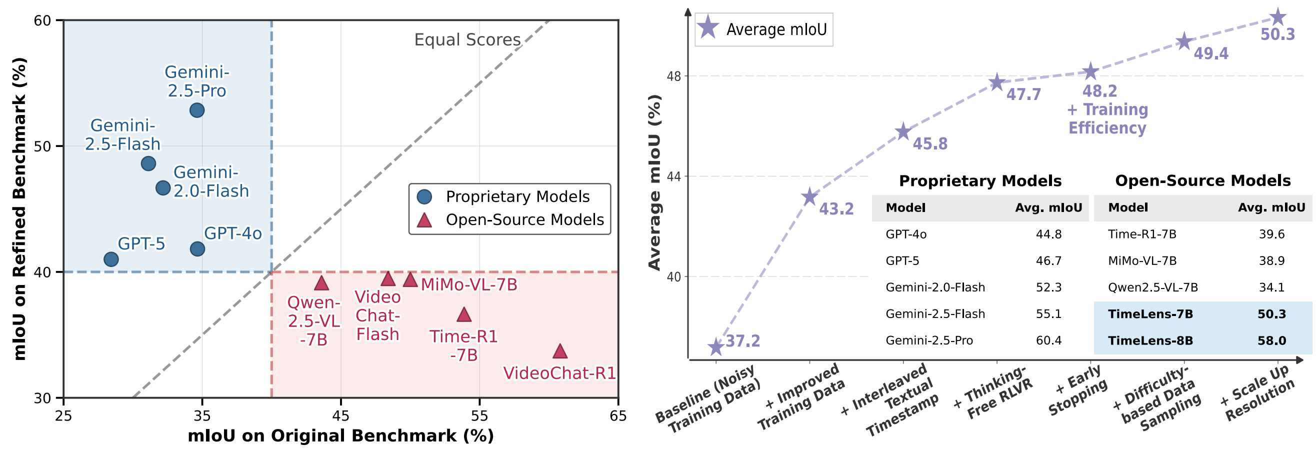


(a) (b)

Figure 2. **(a) Impact of data quality on model evaluation.** A comparison of Mean IoU on original versus our refined Charades-STA benchmarks. The deviation from the diagonal line shows that legacy benchmarks are misleading, as they inflate the results of some open-source models while underestimating proprietary ones. **(b) Cumulative performance gains of TimeLens explorations.** This analysis shows how each component boosts the model's average performance on TimeLens-Bench. From data curation to thinking-free RLVR with early stopping and difficulty-based data sampling, each step demonstrates a clear positive impact towards our final TimeLens model. TimeLens-7B and TimeLens-8B are based on Qwen2.5-VL-7B and Qwen3-VL-8B, respectively.

leading proprietary and open-source models but are also rife with low-quality queries and erroneous timestamps. This noisy data may render current leaderboards misleading and misguide research efforts. To rectify this, we undertook a meticulous data overhaul. We first defined strict criteria for query and timestamp quality, in terms of uniqueness, existence, clarity, and accuracy. We then manually re-annotated three popular datasets (Charades-STA [11], ActivityNet Captions [25], QVHighlights [27]) to create **TimeLens-Bench**, a rigorously cross-validated benchmark. As shown in Fig. 2a, the necessity of this correction is confirmed by a dramatic re-ranking of models on TimeLens-Bench compared to their performance on legacy benchmarks, proving the unreliability of prior evaluation standards. Beyond evaluation, we also fix the noisy training data by automated re-annotation, yielding **TimeLens-100K**, a large-scale, high-quality training dataset.

With our curated data suite as a solid foundation, we conduct in-depth explorations on the algorithmic design principles from three key aspects. First, for timestamp representation, we discover that a simple yet effective interleaved textual encoding strategy outperforms more complex alternatives. Second, we determine that VTG is fundamentally a perception-driven task, and thus employ a pure thinking-free reinforcement learning with verifiable rewards (RLVR) approach that outperforms other training paradigms in both efficiency and performance. Finally, our detailed analysis of RLVR training reveals two key recipes for both performance and training efficiency: (1) early stopping when reward metrics plateau, and (2) difficulty-based data sampling. By integrating these insights and design principles, we ultimately develop **TimeLens models**, a family of MLLMs with superior VTG capability. As shown in Fig. 2b, our model achieves state-of-the-art performance among open-source models and even surpasses proprietary models such as GPT-5 and Gemini-2.5-Flash.

Through these efforts, we identified and addressed long-overlooked quality issues in existing datasets, and derived a series of insights and best practices in algorithmic design. We hope TimeLens can serve as a solid foundation in both data curation and algorithmic design principles, to facilitate future research on building MLLMs with strong VTG capabilities. Our code, data, and models will be open-sourced.

2. Related Work

Temporal Grounding Datasets. Numerous VTG datasets have been proposed, spanning diverse domains [14, 22, 25, 27, 41, 45, 50]. Early works [11, 38, 65] trained and evaluated models on the training and test splits of a single benchmark [25, 45] to assess their ability to fit single-domain data distribution. In recent works [17, 37, 46], large diverse corpuses composed of multiple different source datasets [1, 22, 36, 41, 50, 60] are aggregated for training, and a suite of distinct benchmarks [11, 25, 27] are used to probe the models' real-world cross-domain generalizability.

However, the critical issue of data quality has been overlooked. There lacks a systematic examination on whether existing datasets are reliable enough for training and evaluation. In this paper, we manually inspect existing datasets, identify and correct errors, and produce quality-improved training and evaluation suites for developing more practical

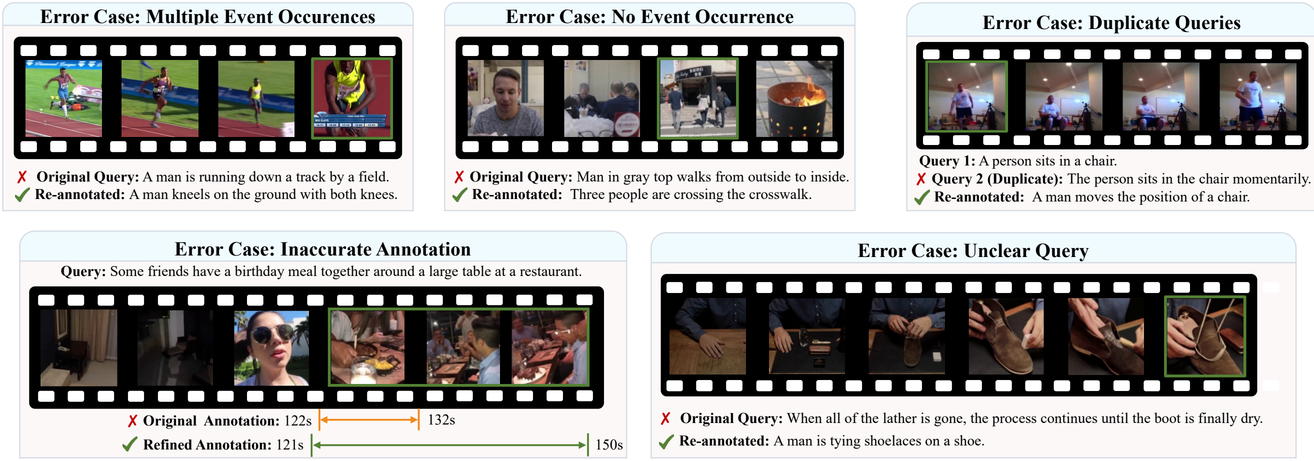


Figure 3. **Qualitative examples of errors and fixes.** We present representative errors identified in existing datasets, spanning different error types, including multiple event occurrences, no event occurrence, duplicate queries for the same video, unclear query, and inaccurate annotation. Through our rigorous manual refinement, these errors have been properly corrected, significantly improving data quality.

VTG models.

MLLMs for Temporal Grounding. Substantial works focus on algorithmic designs to improve MLLMs' VTG capability. One line of research explores model architectures, including token compression methods to reduce computation on long videos [46, 61], timestamp encoding strategies to align the timestamps of each frame with its corresponding features [5, 12, 28, 34, 51, 55, 62]. Another line of works investigate training strategies: introducing various supervised fine-tuning tasks to improve VTG performance [7, 61], or designing verifiable rewards to improve performance via reinforcement learning [4, 33, 54, 59].

Despite the abundance of proposed designs, their inconsistent experimental settings make it difficult to fairly compare their relative merits and establish best practices. In this paper, we systematically analyze these design choices using our quality-assured training and evaluation suites, offering key insights for improving MLLMs' VTG capability.

3. Towards Reliable, High-Quality VTG Data

3.1. Annotation Criteria

Task Formulation. For temporal grounding, a model takes as input a video v and a text query q, localizes the event E described by q, and outputs the corresponding temporal segment $S = (t_{\text{start}}, t_{\text{end}})$. In practice, a video is typically annotated with one or more query-segment pairs $\{(q_i, S_i)\}_{i=1}^{n}$.

Input Criteria. The input video and query should satisfy:

- *Query clarity and specificity*. The query must be clear, precise, and unambiguous for accurate and definitive grounding (A counterexample like "the game continues").
- *Event existence*. The event described in the text query must genuinely exist within the video content.
- *Query uniqueness*. All queries must be unique in a single video. The presence of multiple nearly identical queries describing the same event is equivalent to duplicating or weighting certain samples, leading to biased metrics. Indeed, this issue is severe in Charades-STA dataset.
- *Avoid information leakage in queries*. Queries like "ending credits" leak their temporal position, allowing the model to answer via shortcut, without truly "grounding" the query over the entire video. However, annotators tend to label such queries since they are easy to identify.

Output Criteria. The temporal segment should satisfy:

- *Annotation precision*. The annotated event boundaries should be precise, excluding any subsegments that do not conform to the query's description.
- *Annotation exhaustiveness*. There should be no other time segments outside the annotated one that also satisfy the query's description.

3.2. Manual Auditing and Refinement

We introduce a rigorous and efficient pipeline for auditing and refining existing temporal grounding datasets.

Diagnose-then-Refine. Our pipeline follows a *diagnose-then-refine* workflow. Given a video-query pair from existing datasets, annotators first carefully review the video to identify potential errors against the criteria in Sec. 3.1. If an error is detected, they select the error category, then either revise the query or choose a new valid event to describe. Subsequently, the precise temporal segment is annotated. The core principle is that the **same** annotator performs both error detection and subsequent correction, which not only improves efficiency but also strengthens annotators' awareness of potential errors, thereby reducing the risk of introducing similar ones.

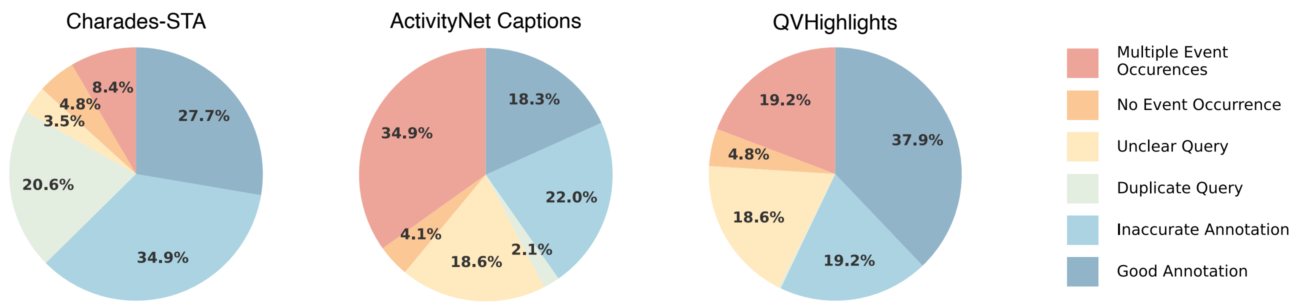


Figure 4. **Statistics of errors indicating alarmingly high proportion of errors in existing datasets.**

Error Identification. Directly applying the abstract criteria from Sec. 3.1 for error detection proves overly challenging for annotators. Therefore, as shown in Fig. 3, we derive from these criteria a set of concrete, easily identifiable error types with clear illustrations. Annotators check whether each error type is present and fill in the corresponding information. Additionally, we group all queries from the same video together to detect violations of "query uniqueness" and improve annotation efficiency. During the process, we do not provide original temporal segments to annotators.

Quality Control. Upon completion of each small data batch, every sample is assigned to a different annotator for cross-validation and error correction. If the error rate in a batch exceeds a threshold, the entire batch is rejected for re-annotation and then validated again. For annotator selection and training, we sampled a small subset of data for trial annotation with over a dozen vendors, then selected the vendor with the highest quality and consistency. Before formal annotation, we provided a detailed handbook and conducted several training sessions. The annotation interface and detailed manual are provided in Sec. G of the appendix.

3.3. Empirical Analysis on TimeLens-Bench

In this section, we present our efforts and findings by applying the above annotation pipeline to existing datasets. We focus on three most widely-used temporal grounding benchmarks: Charades-STA [11], ActivityNet Captions [25], and QVHighlights [27]. These datasets exhibit diversity across video domains, video durations, and query semantics. They are all manually annotated and generally considered the highest-quality VTG datasets available. Therefore, analyzing them offers a representative view of the quality and issues prevalent in existing data. Through diagnosis and refinement, we release **TimeLens-Bench**, comprising refined versions of the three aforementioned benchmarks: Charades-TimeLens, ActivityNet-TimeLens, and QVHighlights-TimeLens. Together, they form a comprehensive evaluation suite that combines diversity with high quality. Detailed statistics for these benchmarks are provided in Sec. B.

Finding 1: Widely-used benchmarks have an alarmingly high proportion of errors.

Error Statistics and Analysis. As shown in Fig. 4, we observe an alarmingly high proportion of errors across different categories in these benchmarks. The distribution of error composition varies across different datasets, yet all datasets exhibit consistently high overall error rates. For example, in Charades-STA, we find that 20.6% of samples violate query uniqueness, while 34.9% exhibit annotation accuracy issues. Such severe errors will lead to unreliable evaluation results and misguide research efforts.

Qualitative Examples of Errors and Fixes. As shown in Fig. 3, various error examples are identified in existing datasets, including multiple event occurrences, no event occurrence, duplicate queries for the same video, unclear query, and inaccurate annotation. Through our rigorous manual refinement, these detected errors have been properly corrected, significantly improving data quality. Our refined datasets provide more reliable evaluation results.

Finding 2: Low-quality evaluation data inflates performance of open-source models, while underestimating proprietary models.

Counter-intuitive Evaluation Results. We evaluate various frontier models on both the original and refined benchmarks, observing *drastically contrasting* performance trends. As illustrated in Fig. 2a, on the original benchmarks, we observe a surprising phenomenon: frontier proprietary models like Gemini-2.5-Pro [8] receive poor scores, whereas open-source models [3, 54] attain significantly higher ones. Conversely, on our refined benchmarks, this trend reverses. The proprietary models exhibit much better results, though with room for improvement, while the open-source models suffer a substantial performance degradation, lagging far behind their proprietary counterparts. This reversal indicates that the original benchmarks produce *misleading* results due to inherent quality flaws, while our refined benchmarks yield results that align more closely with real-world user expe-

Model	Charades-TimeLens				ActivityNet-TimeLens				QVHighlights-TimeLens			
	R1@0.3	R1@0.5	R1@0.7	mIoU	R1@0.3	R1@0.5	R1@0.7	mIoU	R1@0.3	R1@0.5	R1@0.7	mIoU
Proprietary Models												
GPT-4o [23]	60.6	44.5	23.5	41.8	55.2	41.4	25.8	40.4	69.0	54.8	38.5	52.1
GPT-5 [42]	59.3	42.0	22.0	40.5	57.4	44.9	30.4	42.9	72.4	60.4	46.4	56.8
Gemini-2.0-Flash [8]	66.4	53.5	27.1	46.7	62.9	54.0	37.7	49.3	76.2	66.4	48.3	60.8
Gemini-2.5-Flash [8]	68.7	56.1	30.6	48.6	66.8	57.5	41.3	52.5	78.2	69.4	55.0	64.3
Gemini-2.5-Pro [8]	74.1	61.1	34.0	52.8	72.3	64.2	47.1	58.1	84.1	75.9	61.1	70.4
Open-Source Models												
VideoChat-Flash-7B [31]	60.2	37.9	17.8	39.7	35.5	21.8	10.5	24.8	45.2	30.6	16.7	32.7
VideoChat-R1-7B [32]	51.9	30.8	11.7	33.7	35.0	23.9	11.3	25.0	29.3	19.1	9.4	21.5
Time-R1-7B [54]	57.9	32.0	16.9	36.6	44.8	31.0	19.0	33.1	65.8	51.5	36.1	49.2
TRACE [17]	37.2	21.8	9.6	27.1	43.4	33.9	22.0	32.7	49.7	39.1	28.1	39.0
TRACE-uni [17]	38.2	22.9	10.4	28.1	44.3	35.1	22.6	33.6	49.9	40.0	29.2	39.8
TimeSuite [61]	56.3	35.5	18.0	38.1	27.1	17.5	8.6	19.8	27.1	16.9	9.9	21.7
Grounded-VideoLLM [51]	43.3	28.7	13.5	30.0	39.2	29.6	19.5	30.0	43.7	33.8	22.5	33.4
MiMo-VL-7B [9]	57.9	42.6	20.5	39.6	49.3	38.7	22.4	35.5	57.1	42.6	28.4	41.5
Qwen2.5-VL-7B [3]	59.7	37.8	16.6	39.3	44.1	31.0	16.1	31.4	41.5	27.8	15.2	31.6
TimeLens-7B	70.5	55.6	28.4	48.8	62.8	51.0	32.6	46.2	74.1	62.7	43.1	56.0
Qwen3-VL-235B-A22B [2]	71.7	50.8	24.5	47.8	69.0	57.5	39.3	52.2	79.6	70.2	54.5	64.6
Qwen3-VL-8B [2]	69.2	53.4	27.5	48.3	62.1	51.2	34.4	46.8	74.2	64.6	49.3	59.4
TimeLens-8B	**76.6**	**63.0**	**35.2**	**55.2**	**68.9**	**58.4**	**40.6**	**53.2**	**80.2**	**71.6**	**55.5**	**65.5**

Table 1. **Main Results.** We benchmark the performance of various state-of-the-art proprietary and open-source models on TimeLens-Bench. Our TimeLens models are built upon their respective baseline models (preceding rows in the table). Our **TimeLens-7B** not only delivers substantial improvements over the Qwen2.5-VL baseline but also closes the gap with the more powerful Qwen3-VL-8B model. Building upon the stronger Qwen3-VL baseline, our **TimeLens-8B** pushes performance even further, setting a new state-of-the-art among open-source models and surpassing prominent proprietary models like GPT-5 and Gemini-2.5-Flash.

rience, providing reliable evaluation for developing better VTG models.

3.4. Training Data Re-annotation

By applying our manual pipeline from Sec. 3.2 to a sampled subset of existing VTG training corpus [1, 22, 41, 50, 60], we found that the training data exhibits an even higher error rate compared to the evaluation benchmarks. This motivated us to refine training data based on scalable re-annotation. Given the vast scale of the training sets, we employ an automated pipeline to improve their quality based on advanced multimodal models. Owing to the poor quality of these training datasets, especially the high proportion of queries that fail to meet our criteria in Sec. 3.1, we re-annotate the videos rather than refining existing labels. Through this process, we curate **TimeLens-100K**, a large-scale, high-quality, and diverse VTG training set. Additional details are provided in Sec. H.

> *Finding 3:* Improved annotation quality in training data yields stronger grounding ability.

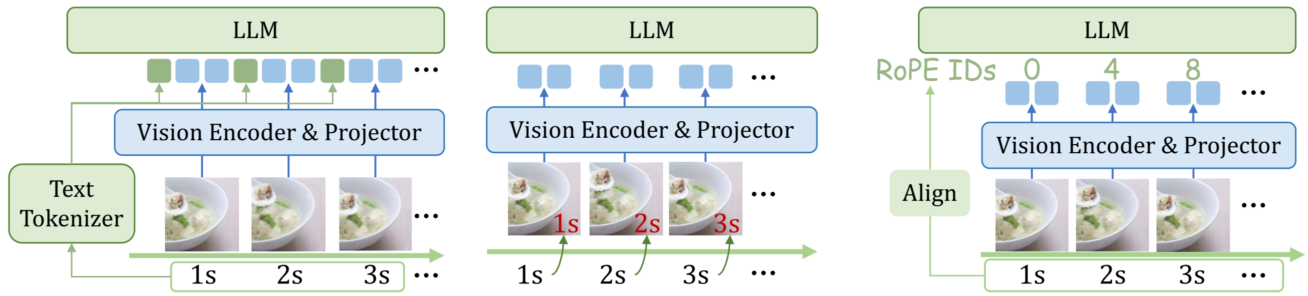
As presented in Fig. 2b, models trained on TimeLens-100K demonstrate substantially improved performance on our refined evaluation benchmarks. This performance gain serves as a direct validation of the data's enhanced quality. Notably, our automated re-annotation for training data is developed entirely independently of the manual benchmark refinement process, ensuring an unbiased evaluation.

4. Benchmarking Grounding MLLMs

In this section, we benchmark the performance of various state-of-the-art proprietary and open-source models on TimeLens-Bench, including our TimeLens models derived from the exploration in Sec. 5.

Evaluation Metrics. We evaluate VTG performance using the "R1@m" metric, which measures the proportion of test instances where the highest-ranked predicted segment achieves an IoU exceeding threshold m (where m takes values from 0.3, 0.5, 0.7). Additionally, we employ mIoU as a primary measure, computing the mean IoU across the entire test set for conciseness.

Evaluation Results. As shown in Tab. 1, we observe a significant performance gap between existing open-source and proprietary models, and our TimeLens models substantially narrow this gap. TimeLens-7B delivers substantial improvements over its baseline, demonstrating the effectiveness of the insights and best practices obtained from our experiments in Sec. 5. It surpasses strong open-source com-



(a) Interleaved Textual Timestamp Encoding. (b) Visual Timestamp Overlay. (c) Position-embedding-based Time Encoding.

Figure 5. **Illustration of different timestamp encoding schemes.** (a) **Textual Encoding** uses the text tokenizer of LLMs to tokenize textual timestamps into textual tokens. (b) **Visual Overlay** directly overlays timestamps as visual text onto the corresponding frames. (c) **Position-embedding-based Methods** aligns the positional encodings of visual tokens in the LLM with the sampling time of each frame.

Method	Timestamp Format	Charades-TimeLens				ActivityNet-TimeLens				QVHighlights-TimeLens			
		R1@0.3	R1@0.5	R1@0.7	mIoU	R1@0.3	R1@0.5	R1@0.7	mIoU	R1@0.3	R1@0.5	R1@0.7	mIoU
Position Embed. [3]	-	57.9	32.0	16.9	36.6	44.8	31.0	19.0	33.1	65.8	51.5	36.1	49.2
Visual Overlay	Frame Index	65.5	48.0	22.2	44.0	47.5	34.0	17.8	33.3	61.4	43.6	24.1	42.3
	Raw Timestamp	67.6	50.7	25.7	46.3	54.0	42.2	26.3	39.8	70.0	58.3	42.1	53.6
Not-Interleaved Textual Prefix	Raw Timestamp	64.9	49.4	27.3	45.8	48.2	35.5	21.4	35.2	59.7	45.8	26.4	42.8
Interleaved Textual Prefix	Frame Index	66.0	51.6	25.6	45.6	51.0	39.1	23.1	36.9	64.4	52.1	32.3	47.2
	Raw Timestamp	**70.0**	**53.9**	**28.1**	**48.3**	**57.9**	**46.3**	**30.5**	**43.1**	**73.0**	**62.2**	**46.1**	**56.7**

Table 2. **Ablation on timestamp encoding methods.** For each method, we experiment with two timestamp formats: raw timestamps (*e.g.*, "10.2s") or frame indices (*e.g.*, "1, 2, 3"). "Position Embed." means "Position Embedding". Results show that **interleaved textual prefix** with raw timestamps is the most effective approach, while maintaining simplicity.

petitors such as Time-R1-7B [54] and MiMo-VL-7B [9], as well as proprietary models like GPT-4o [23] and GPT-5 [42]. More remarkably, on the already stronger baseline Qwen3-VL-8B, our TimeLens-8B model still achieves substantial performance gains, establishing a new state-of-the-art among open-source models and even surpassing frontier proprietary models like Gemini-2.5-Flash [8].

5. Exploring Algorithmic Designs

In this section, we conduct a systematic study on the algorithmic designs for improving MLLMs' VTG performance, covering various aspects from model architectures to training strategies. Leveraging our high-quality training and evaluation suites as a reliable testbed, we derive several novel and valuable insights. As shown in Fig. 2b, each of our findings contributes a non-trivial performance gain, ultimately culminating in our **TimeLens** model.

Experimental Setup. Our experiments use Qwen2.5-VL-7B [3] as the baseline. For RLVR experiments, we employ GRPO [47] as optimization method. We use TimeLens-Bench for evaluation and TimeLens-100K for training. To ensure rigor, all ablation studies are based on the final, best-performing model configuration, isolating the impact of a single design choice at a time. Due to limited computational resources, we adopt a lower per-frame resolution for our ablation experiments. More implementation details are provided in Sec. C of the appendix.

5.1. Timestamp Encoding

Finding 4: Encoding timestamps as *interleaved textual prefix* is the most effective while maintaining simplicity.

To enable MLLMs to perform temporal grounding, a critical design decision is timestamp encoding (*i.e.*, aligning the timestamp of each frame with its corresponding features). Effective timestamp encoding allows the model to accurately perceive the absolute temporal position of each frame and the relative order between frames, thereby producing precise localization results. As illustrated in Fig. 5, various timestamp encoding strategies have been proposed:

- *Position-embedding based.* These methods adapt position embeddings in LLMs to represent the temporal position of each frame. For example, MRoPE [3, 9] and 3D RoPE [48] extend pure-text RoPE to multimodal scenarios, encoding the spatial and temporal dimensions of video frame tokens.

Training Paradigm	Training Time	Charades-TimeLens				ActivityNet-TimeLens				QVHighlights-TimeLens			
		R1@0.3	R1@0.5	R1@0.7	mIoU	R1@0.3	R1@0.5	R1@0.7	mIoU	R1@0.3	R1@0.5	R1@0.7	mIoU
SFT (32K Data)	1.0×	68.8	53.0	26.2	47.4	53.3	42.6	27.5	39.9	65.8	54.8	40.6	52.0
SFT (100K Data)	2.4×	70.6	54.9	27.1	48.6	53.2	43.1	27.2	39.7	63.1	51.1	36.9	49.0
Thinking-based RLVR	1.9×	60.3	46.4	24.7	42.7	54.3	44.2	29.1	41.2	72.1	**62.7**	**48.2**	**57.8**
SFT + Thinking-free RLVR	2.9×	**71.7**	**56.7**	**29.8**	**50.1**	56.9	46.1	30.1	42.7	72.2	60.6	43.8	55.9
Thinking-free RLVR	1.0×	70.0	53.9	28.1	48.3	**57.9**	**46.3**	**30.5**	**43.1**	**73.0**	62.2	46.1	56.7

Table 3. **Ablation on different training paradigms.** We compare the performance and efficiency of different training paradigms, showing that thinking-free RLVR achieves the best performance while maintaining high efficiency. All training is conducted on our quality-improved TimeLens-100K training data. Training time is measured on 8× H20 GPUs, where 1.0× corresponds to approximately 4h10m. As described in Sec. 5.3, before RLVR training, offline inference on the training data is required to select samples with appropriate difficulty; this time is also included in the reported RLVR training time.

- *Visual overlay.* These methods [7, 12, 55] directly overlay timestamps or frame index onto each frame, enabling MLLMs to "read" the temporal position through their OCR capabilities.
- *Textual encoding.* These methods convert timestamps into text tokens using the MLLM's text tokenizer. There are two main variants: the *Interleaved* approach [5, 15, 20, 34, 56] in Fig. 5a inserts timestamp tokens before the visual tokens of each frame. In contrast, the *Non-interleaved* approach [29, 31, 52] adds an instruction like "This video samples N frames of a T-second video at $t_1, t_2, \ldots$ seconds." into the prompt.

We conduct a comprehensive comparison of different timestamp encoding methods. For each method, we experiment with two timestamp formats: raw timestamps (*e.g.*, "10.2s") or frame indices (*e.g.*, "1, 2, 3"), which are simpler but neglects the temporal interval between frames. As shown in Tab. 2, our results reveal: Position-embedding based methods yield unsatisfactory results. Given that they require fundamental modifications to the RoPE mechanism in LLMs, their practicality is limited without large-scale retraining. Instead, interleaved textual prefix with raw timestamps achieves the best performance among all approaches, while remaining simple and intuitive.

5.2. Optimization Paradigms

> ***Finding 5:*** For the optimization paradigm, a pure thinking-free RLVR approach achieves superior performance and efficiency. Both SFT and thinking-based RLVR are not necessary.

In this section, we review different training paradigms and conduct systematic experiments to compare their effectiveness and efficiency for VTG, seeking insights into the optimal training paradigm.

Earlier works [17, 18, 22, 46, 61] employ supervised fine-tuning (SFT) to improve MLLMs' VTG capability. Recently, some works [4, 54] utilize reinforcement learning with verifiable rewards (RLVR), following a "think-then-answer" approach [16] (details in Sec. C): during sampling, the model first generates an explicit thinking process and then produces the final answer. The task-specific VTG accuracy reward is computed only on the final answer. Despite these efforts, there lacks a systematic comparison of the respective merits of these methods, leaving some key questions unanswered:

- **Is RLVR superior to SFT?** While the pioneering work Time-R1 [54] demonstrates that RLVR outperforms SFT, they compare the two methods using the same amount of training data, despite RLVR requiring significantly more training time. A fair comparison under equal training budgets remains absent.
- **Is explicit "thinking" necessary for RLVR?** Recent works suggest that the thinking process is not essential when applying RLVR to visual perception such as counting [30, 43]. Whether this holds for VTG, a predominantly perception-oriented task, remains unanswered.
- **Does a preceding SFT phase benefit RLVR?** An SFT phase prior to RLVR is typically employed to enhance the model's capability and facilitate subsequent RLVR training [9, 48]. However, whether this preceding SFT phase actually improves final performance in the VTG scenario remains unexplored.

In Tab. 3, we compare the performance and efficiency of different training paradigms. Our results reveal that thinking-free RLVR surpasses both SFT and thinking-based RLVR in performance while being more efficient. Adding a preceding SFT phase before RLVR yields no significant performance gain compared to pure RLVR. Overall, a pure thinking-free RLVR approach maintains simplicity, superior performance, and high efficiency.

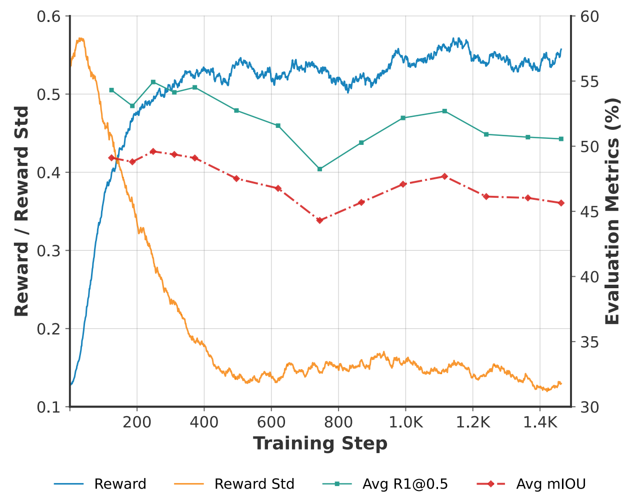


Figure 6. **The effectiveness of early stopping for RLVR.** We show the trends of training reward and evaluation metrics during RLVR training. When the temporal IoU reward and the within-group reward standard deviation plateau, performance reaches its peak. Continued training beyond this point leads to performance degradation. Therefore, performing *early stopping* when the reward plateaus ensures optimal training efficiency and performance. Training is conducted on ∼12K samples selected from TimeLens-100K via difficulty-aware sampling.

5.3. Recipes for RLVR Training

Building on the finding in Sec. 5.2 that thinking-free RLVR is the optimal training paradigm, in this section, we further explore effective recipes for RLVR training, focusing on two key questions: (i) How long should we train? (ii) How to effectively sample training data?

> ***Finding 6:*** For RLVR training, performing early stopping when reward metrics plateau saves computational cost, while preventing performance degradation.

How long should we train? In SFT, the prevailing wisdom is "train longer, generalize better" [19]. Given training data with sufficient scale and quality, we typically train MLLMs for at least one full epoch over the entire dataset, ensuring the model sees as much data as possible to enhance generalization. However, whether this strategy is optimal for RLVR remains to be explored.

In Fig. 6, we conduct RLVR training on ∼12K data from TimeLens-100K, tracking the reward and evaluating model checkpoints at different training steps on our evaluation benchmarks. When the temporal IoU reward and the within-group reward standard deviation plateau, model performance has reached its peak. Continued training beyond this point leads to performance degradation. Therefore, in RL training, even with sufficiently high data quality, training for a full epoch over all available data is suboptimal. A good practice is performing *early stopping* when reward metrics plateau, which not only saves computational cost but also prevents performance degradation.

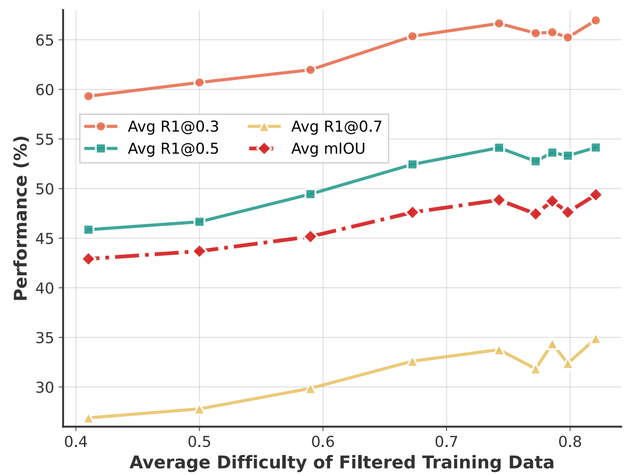


Figure 7. **The importance of difficulty-based training data sampling for RLVR.** We investigate the impact of training data with different difficulty levels on performance by adjusting the mean of the Gaussian distribution for difficulty-based data sampling. Model performance improves as the average sample difficulty increases, and eventually plateaus when difficulty becomes high. This demonstrates that selecting samples with sufficiently high difficulty is crucial for achieving optimal performance.

> ***Finding 7:*** For RLVR training, sampling training data with sufficiently high difficulty relative to the model is crucial for performance.

How to sample training data? For RLVR training, it is crucial to select samples with appropriate difficulty relative to the model, and many works propose assessing training sample difficulty and employing difficulty-aware sampling [21, 54, 57]. To evaluate the impact of sample difficulty on video temporal grounding, we conduct experiments on our TimeLens-100K high-quality training corpus. Following prior works [54, 57], we use the model to be trained to perform offline inference on the training data, compute IoU metrics to estimate sample difficulty, and then perform Gaussian sampling based on sample difficulty (details in Sec. C). By varying the mean of the Gaussian distribution, we obtain training sets with different difficulty levels relative to the model, and conduct RLVR training on each set independently.

As shown in Fig. 7, model performance improves as the average sample difficulty increases, and eventually plateaus when difficulty becomes sufficiently high (over 0.75). This trend demonstrates that selecting training samples with sufficiently high difficulty relative to the model is crucial for achieving optimal performance.

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